

Kim Yungju

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EDUCATION

NATIONAL UNIVERSITY OF SINGAPORE (NUS)

Aug 2024 - Present

Bachelor of Computing (Computer Science), Honours

- Transferred from BBA & Economics to Computer Science to pursue technical specialization
- Expected Graduation: December 2027

EXPERIENCE

LUMCLOON ENERGY | Data Center Systems Intern

Dec 2025- Jan 2026

- Authored a 46-page internal assessment on Ireland's new Large Energy User (LEU) framework, analyzing the transition of the data center market into a regulatory-driven maturity stage.
- Modeled grid capacity constraints and de-rating mechanisms for hyperscale deployment, evaluating the integration of SOFC (160MW), BESS (250MW), and CPPA renewables.
- Identified a 73MW capacity deficit by applying CRM de-rating curves and formulated a Credible Plan for 80% renewable matching, securing the technical foundation for a compliant grid connection application.

REAL COACHING | Software Engineer Intern

May 2025 - August 2025

- Built an internal client management dashboard using Next.js, PostgreSQL, and Shadcn UI to track session schedules, client progress, and coach assignments, replacing manual spreadsheet tracking and reducing administrative overhead by 30%.
- Created reporting and admin tools with Next.js, PostgreSQL, and Shadcn UI to aggregate coaching engagement metrics and manage program content, supporting data-driven decisions and streamlining onboarding workflows.

CO-CURRICULAR ACTIVITIES

Developer Group @ NUS Computing | Head of Technology

Aug 2025 - Present

- Contributed to a student-led tech community (2,300+ followers) connecting communities with technology.
- Designed and developed the organization's website to showcase events, initiatives, and partner collaborations, improving visibility and outreach.
- Organized and managed the technical operations of Hack4Good 2026, a flagship hackathon designing impactful solutions for non-profit organizations.

PROJECTS

PriceWise — AI Shopping Agent

- Developed an autonomous product research agent that compares prices and reviews across retailers, delivering structured receipts via a conversational interface with selective human-in-the-loop approval.
- Engineered a LangGraph agent with 10 orchestrated tools, dual-mode SSE streaming, persistent checkpointing via AsyncPostgresSaver, and conversation summarization with safe message splitting.
- Built with LangGraph, FastAPI, OpenAI, Tavily, Pydantic, PostgreSQL, Next.js, and Docker, deployed on Railway (backend) and Vercel (frontend).

InterviewPilot — AI Mock Interview Platform

- Built a full-stack mock interview platform that generates role-tailored questions, conducts speech-based interviews with webcam recording, and delivers AI-scored feedback.
- Implemented multi-turn conversational flow with AI-generated follow-up questions, fire-and-forget video upload pipeline with late URL patching, and cross-browser MediaRecorder MIME negotiation.
- Built with Next.js, React, TypeScript, OpenAI, Drizzle ORM, Supabase, and Clerk, with bilingual support (English/Korean) including dual TTS pipeline and CJK PDF export.

Castory — AI Podcast Platform

- Developed a full-stack SaaS platform enabling users to generate AI-narrated podcasts from trending news via a 5-step guided workflow using OpenAI.
- Implemented chunked TTS pipeline, persistent audio player, full-text search, draft auto-save, and dynamic user profiles with listener analytics.
- Built with Next.js, React, TypeScript, Convex (serverless backend + real-time DB), and Clerk (webhook-synced auth), deployed on Vercel.

SKILLS AND OTHER INFORMATION

- Fluent in written and spoken English and Korean
- Programming Languages: JavaScript, Java, Python, and SQL
- Frameworks & Tools: Next.js, React, LangGraph, FastAPI, Docker, PostgreSQL, Supabase